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Genetic estimates of population structure and dispersal in a benthic stream fish

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Abstract– Comparing genetic and demographic estimates of dispersal in freshwater fish can improve understanding of movement distributions and population connectivity. Here we examined genetic variation among mottled sculpin (*Cottus bairdi*) in the Nantahala River (North Carolina, USA) to compare genetic estimates of dispersal with estimates derived from mark–recapture studies of individual movement. Microsatellite-based analysis of gene flow revealed evidence of strong isolation by distance among locations spanning only 5.6 km and limited dispersal among clusters of sites separated by swift cascades. Estimates of between-cluster contemporary dispersal rates derived from Bayesian assignment tests ranged from 1% to 6%, with most movement occurring among adjacent clusters in a downstream direction. Evidence of a long-term net immigration asymmetry and decreasing genetic diversity from downstream to upstream locations indicates that historical patterns of stream colonisation contrast with contemporary dispersal patterns. Our findings are largely consistent with predictions from individual movement patterns but suggest that long moves (>500 m) are more frequent, and maximum dispersal distances are greater than what has been reported from mark–recapture studies. The discrepancy may reflect spatial limitations of mark–recapture methods or temporal variation in dispersal in individuals and populations.

Key words: *Cottus bairdi*; mottled sculpin; population connectivity; isolation-by-distance; mark–recapture

Introduction

By moderating the size and structure of local populations, dispersal can enable the persistence of species in heterogeneous environments, including those affected by anthropogenic habitat loss or fragmentation. This is particularly true for fishes inhabiting river-stream networks where dispersal is required to secure spatially or temporally patchy resources. Yet, understanding of freshwater fish dispersal is limited because direct demographic measurement at ecologically relevant spatial scales is technically difficult. Indirect genetic measurement can offer complementary information on dispersal, and there is growing interest in combining demographic and genetic approaches to potentially obtain information that neither approach can provide alone. Few comparative studies have been carried out, however, so little guidance is available on how to interpret and best make use of information from either genetic approaches or a combination of approaches.

Demographic and genetic approaches do not necessarily result in comparable or congruent estimates of dispersal. Mark–recapture methods, for example, may underestimate dispersal potential by truncating movement distributions (Koenig et al. 1996; Rodriguez 2002; Morales et al. 2004). Movement across long distances is notoriously difficult to assess, particularly when it occurs during several life-history stages (Morrissey & Ferguson 2011). Genetic inferences of dispersal, on the other hand, incorporate rare events that can define movement distributions, but may not reflect contemporary ecological conditions (Whitlock & McCauley 1999). Inferences derived from genetic indices therefore can be imprecise or can vary greatly among different methods of analysis, particularly

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when population genetic differentiation is weak or absent (Wilson et al. 2004).

An increasing number of empirical studies are comparing demographic and genetic approaches to determine the conditions that result in consistent, complementary or incongruent information on dispersal (Stevens et al. 2006; Gauffre et al. 2008; Lowe et al. 2008; Pearce et al. 2008; Purrenhage et al. 2009; Morrissey & Ferguson 2011; Safner et al. 2011). Comparisons involving stream fish have so far largely focused on salmonids, such as the mark–recapture and microsatellite-based genetic studies of brook charr (*Salvelinus fontinalus*) that have found evidence of incongruent estimates among sites within a watershed (Wilson et al. 2004) but congruent estimates among sites across watersheds (Adams & Hutchings 2003). In a related study, Morrissey & Ferguson (2011) were able to resolve discordant estimates of brook charr dispersal by accounting for movement across life-history stages within and among individuals. Determining whether findings from these and other studies of salmonids and game species (Miller et al. 2001) are broadly representative of freshwater fish, however, requires conducting additional comparisons involving other ecologically important species.

Mottled sculpin (*Cottus bairdi*) have proven to be an exceptionally useful ecological model for studying freshwater fish dispersal. Mottled sculpin are a widely distributed benthic species that can dominate cold-water fish assemblages across temperate streams of North America (Freeman et al. 1988; Grossman et al. 2002). The demography and movement of mottled sculpin have been extensively examined in several regions of the United States (Petty & Grossman 2004; Lamphere 2005, Breen et al. 2009). Populations tend to be stable over time (Freeman et al. 1988; Grossman et al. 1998; Rashleigh & Grossman 2005), with high residency and low rates of emigration (Petty & Grossman 2010, 2004; Fiumera et al. 2002; Lamphere 2005). Larger individuals (greater or equal to 40 mm) are exceptionally sedentary. Mean displacement distances as low as 4.4 m have been observed in short-term, fine-grain mark–recapture studies (Petty & Grossman 2004). However, mean displacement distances up to an order of magnitude greater (45–65 m), maximum movement distances up to 511 m, and evidence of movement bias have all been reported (Lamphere 2005; Breen et al. 2009).

Despite extensive use of genetic methods for assessing reproductive strategies and success (Kinziger & Raesly 2001; Fiumera et al. 2002; Jones et al. 2007), no genetic studies have been completed on movement patterns and dispersal distances of mottled sculpin (though see the study by Howard & Morgan 1993). Observational estimates of population stability and individual movement suggest that genetic approaches can offer valuable insight into movement potential, dispersal rates and directional bias (Whitlock & McCauley 1999). Evidence of restricted movement and stable populations, for example, suggests that genetic approaches may yield reasonable estimates of migration rates from allele frequency indices based on assumptions of equilibrium (Slatkin & Barton 1989; Neigel 1997; Whitlock & McCauley 1999). Evidence of variation in growth and survival according to habitat quality (Rashleigh & Grossman 2005) and low emigration rates (Petty & Grossman 2004) suggests that discrete population structure may arise because of patchy distributions or in-stream barriers to movement, even over extremely fine spatial scales. Therefore, genetic differentiation is likely sufficient to identify recent immigrants and to assign populations of origin from individual-level comparisons of multilocus genotypes (Beerli & Felsenstein 1999; Cornuet et al. 1999; Morrissey & Ferguson 2011). Inferences of movement potential and direction may also capture rare events, as evidence of upstream bias and frequent long distance movements (e.g., 16% of individuals moving more than 100 m per year) raises the possibility that movement among populations could be higher than direct methods can readily detect, and that exchange among populations is directional (Lamphere 2005; Breen et al. 2009).

In this study, we assessed population structure and dispersal of mottled sculpin from patterns of genetic variation. Our objectives were to estimate dispersal distances, migration rates and movement bias. This involved characterising geographical patterns of genetic differentiation among mottled sculpin over fine spatial scales, and comparing the rate and direction of exchange among sample sites according to longitudinal location and potential barriers to dispersal. Meeting these objectives enabled us to evaluate how genetic estimates of dispersal and population connectivity correspond to inferences drawn from mark–recapture studies of individual movement.

Materials and methods

Study location

In summer 2003, specimens of *C. bairdi* were collected from eight locations (Sites A–H) across a 5.6-km headwater reach (35° 04′ 40″ N, 83° 32′ 05″ W) of the upper Nantahala River in western North Carolina (USA). Sites were distributed across the length of the reach, encompassing heterogeneous in-stream conditions. Sites were spaced over a broad distribution of pairwise distances, with distances between proximate sites varying between 250 m and over 1 km (Fig. 1, Table 1). Site C was located 200 m

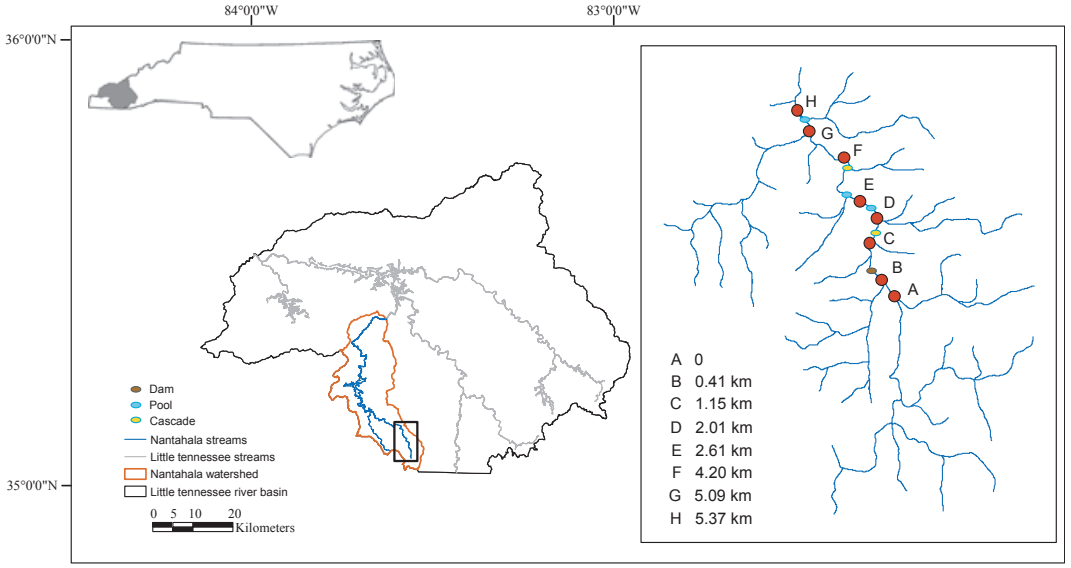

Fig. 1. The position of sampling sites and potential migration barriers within the Nantahala River in the Little Tennessee River basin in North Carolina (USA). Distances between sites provided in table, with Site A as the reference.

upstream, and Site F was located 100 m downstream from swift cascades. Deep pools were present 100 m upstream and 200 m downstream of Site E, and a 1-m-high log dam was present just downstream of Site B (Fig. 1).

The study area falls within a forested section of the drainage basin, with mature and intact riparian vegetation composed of a mixed hardwood canopy and an understory of rhododendron (*Rhododendron maximum*) and mountain laurel (*Kalmia latiflora*). The sampled sites were located between 1050 and 1080 m above sea level, held cold water (with summer temperatures averaging 16 °C) and exhibited a maximum width of 12 m (Lamphere 2005). The fish assemblage in the study area was dominated by mottled sculpin, rainbow trout (*Onchorhynchus mykiss*), brown trout (*Salmo trutta*) and brook trout (*Salvelinus fontinalis).* Blacknose dace (*Rhinichthys atratulus*) and creek chub (*Semotilus atromaculatus*) also were found, but were less abundant than mottled sculpin and trout. Mottled sculpin outnumbered trout by five to one, and adult trout were limited to pools over 0.4 m deep (Lamphere 2005). Crayfish and *Desmognathus* salamanders were also conspicuous residents.

Table 1. Latitude and longitude of sample sites; sample sizes (*N*); numbers of male, female, and juvenile individuals (small individuals with uncertain sexual characteristics); minimum total length [TL (MIN)]; maximum total length [TL (MAX)]; total number of alleles per site and cluster (Alleles); heterozygosity and private allele frequency within sites and within clusters.

										Heterozygosity		Private allele frequency	
Locations	Latitude (W)	Longitude (N)	*N*	Males	Females	Juveniles	TL (MIN)	TL (MAX)	Alleles	Mean	SE	Mean	SE
Site													
A	35 02 34.20	83 30 30.27	35	14	16	5	51	115	96	0.556[a]	0.022	0.0013[a]	0.0013
B	35 02 40.55	83 30 36.39	39	12	20	7	52	102	97	0.598[a]	0.025	0.0012[a]	0.0012
C	35 03 07.22	83 30 48.42	42	15	18	9	49	105	108	0.559[a]	0.022	0.0035[a]	0.0026
D	35 03 24.84	83 30 44.09	39	12	19	8	55	107	105	0.588[a]	0.02	0.0033[a]	0.0023
E	35 03 34.58	83 30 59.41	30	6	8	16	46	91	102	0.578[a]	0.033	0.0047[a]	0.0034
F	35 04 03.64	83 31 15.26	43	11	7	25	49	85	115	0.631[b]	0.023	0.0060[a]	0.0028
G	35 04 21.89	83 31 42.61	40	12	23	5	53	108	107	0.613[ab]	0.026	0.0021[a]	0.0015
H	35 04 30.87	83 31 49.00	44	14	26	4	56	109	116	0.647[b]	0.02	0.0069[b]	0.0034
Cluster													
ABC			116	41	54	21	51	107	122	0.571[a]	0.013	0.00282[a]	0.0031
DE			69	18	27	24	51	99	120	0.583[a]	0.018	0.0039[a]	0.0038
FGH			127	37	56	34	53	101	139	0.631[b]	0.013	0.017[b]	0.0075

SE, standard error.

Superscript letters indicate statistically distinct groups (*t*-tests, $P < 0.05$ after Bonferroni correction). Mean values for TL (MIN), TL (MAX), and total number of males, females and juveniles are presented for clusters of sites.

The study reach is confluent with three tributaries (Hurricane Creek, Big Indian Branch and Bearpen Creek), where a multiyear, mark–recapture study and a manipulative experiment were being conducted on mottled sculpin (Lamphere 2005). The mark–recapture surveys recovered a leptokurtic movement distribution exhibiting an upstream bias. The surveys also found that 48% of recaptures moved 20 m or more, and that longer moves varied seasonally. Individuals moved more often, and farther, from mid-May to autumn than during winter or the spring breeding season. Observational surveys and field manipulations both demonstrated that *C. bairdi* use conspecific density as a direct cue for reach-scale movement rather than prey density, which is a proximal cue for fine-scale (<2 m) movement. Individuals moved towards sites with low conspecific density and favourable nesting habitat in spring and towards sites with high density and favourable foraging habitat in summer.

Specimen collection, DNA extraction and microsatellite genotyping

Specimens were collected from each site by repeat-pass electrofishing of a 30-m reach with habitat favoring high densities of mottled sculpin. From 30 to 49 mottled sculpin were sampled at each location. A tissue clip was taken from the caudal fin of each individual. All fin clips were immediately preserved in 95% ethanol. Individuals were measured for length and, when possible, sexed before being released.

Tissues were first digested overnight in a proteinase K solution, after which DNA was extracted following a 25:24:1 phenol:chloroform:isoamyl alcohol protocol. Twelve microsatellite loci were then amplified for all samples. Two loci (Cb14 and Cb51) were developed for *C. bairdi* (Fiumera et al. 2002), and the other ten loci (Cgo5, Cgo18, Cgo33, Cgo34, Cgo42, Cgo56, Cgo310, Cgo1033 and Cgo1114) were developed for *C. gobio* (Englbrecht et al. 1999). Each PCR consisted of a 15-μl mixture that included 10–50 ng of DNA, 1 U of *Taq* polymerase (Invitrogen, Carlsbad, CA, USA), 75 μM of each dNTP, 1 pmole of each primer, 1× PCR buffer (200 mM Tris–HCl, pH 8.4; 500 mM KCl) and 2 mM $MgCl_2$. PCR products were generated following a 'touchdown' thermal regime of an initial denaturing step of 95 °C for 60 s, followed by 12 cycles of 95 °C for 30 s, 64 °C for 30 s (with a decrease in temperature of 0.8 °C per cycle) and 72 °C for 90 s. This was followed by 23 cycles of 95 °C for 30 s, 54 °C for 30 s and 72 °C for 90 s, followed by a final extension stage of 72 °C for 15 min. The forward primer of each primer pair was labelled with either a HEX, 6-FAM or a TET fluorescent dye, so that PCR of each locus generated labelled amplicons for sizing the loci against a CST ROX 50–500 size standard (BioVentures, Inc., Murfreesboro, TN, USA) on multiplexed acrylamide gels run on a MJ Basestation (Biorad, Hercules, CA, USA). The resulting gel files were analysed using *cartographer*® software (Biorad).

Analysis of genetic variation and population structure

We used GENEPOP v4.0.11 (Raymond & Rousset 1995) to identify departures from Hardy–Weinberg equilibrium (HWE) for each locus and to apply Fisher's exact tests for detecting linkage disequilibrium among loci. GENEPOP v4.0.11 was also used to calculate allele frequencies, observed and expected heterozygosity, and to identify private alleles. Sequential Bonferroni corrections were implemented to account for simultaneous comparisons when determining statistical significance (Rice 1989).

Population subdivision was assessed by calculating pairwise values for linearised F_{st} (Slatkin 1995) in ARLEQUIN v3.5 (Excoffier et al. 2005). We also calculated and bootstrapped (based on 1000 replicate calculations) the variance in the proportion of shared alleles (D_{ps}) across sample locations using MICROSATELLITE ANALYZER software (Bowcock et al. 1994; Dieringer & Schlötterer 2002), and constructed a neighbour-joining tree based on these calculations using the 'Neighbor', 'Consense' and 'Drawtree' subroutines of PHYLIP v3.63 (Felsenstein 1989, 2004).

A matrix of pairwise F_{st} values was compared against a corresponding matrix of between-site geographical distances with a Mantel's test (Mantel 1967) to determine whether genetic variation among mottled sculpin in the study reach corresponds to a pattern of isolation by distance (IBD). Mantel's tests were performed in FSTAT v2.93 (Goudet 1995), with statistical significance estimated from comparison of the observed data to 10,000 permutations. The nlmod package for R (R Development Core Team 2005) was used to construct a spline correlogram (Bjornstad & Falck 2001) to identify significant genetic autocorrelations with geographical distances among sites.

Analysis of migration

Assignment tests were conducted for comparison to estimates of population subdivision derived from pairwise F_{st} values and to identify recent migrants among populations (Hänfling & Weetman 2006). Estimates were based on Bayesian models implemented in STRUCTURE v2.1 (Pritchard et al. 2000) and BAYESASS v1.3 (Wilson & Rannala 2003). While both approaches estimate the genetic identity of each individual, STRUCTURE v2.1 assignments reflect populations conforming to HWE conditions, whereas HWE is not a requirement for assignment testing in BAYESASS v1.3. Accordingly, loci that consistently

exhibited departures from HWE were not used in analyses with STRUCTURE v2.1 but were included in analyses with BAYESASS v1.3. After several trial runs, we chose a burn-in period of 30,000 Markov Chain Monte Carlo (MCMC) iterations and collected data from an additional 10^6 iterations for five replicate sets of runs in STRUCTURE v2.1 with K (the number of populations) iteratively set from one to eight. The number of populations within the study area was identified based upon the greatest P(X | K) value and log-likelihood tests comparing approximate values. An analysis of molecular variance (AMOVA) also was performed in ARLEQUIN v3.5 (Schneider et al. 2000) with groups reflecting estimates from STRUCTURE v2.1. BAYESASS v1.3 was used to determine the number of first-generation migrants (m_1) and the directionality of exchange among individual locations and groups estimated by STRUCTURE v.2.1 (Hänfling & Weetman 2006). Individuals with genotypes assigned to a different site than the site of capture were identified as first-generation migrants from other sites. Estimates of m_1 were obtained from the averages of five replicate runs of 5×10^7 iterations following 3×10^7 burn-in iterations. Individuals identified as likely offspring of migrants were considered residents.

We estimated long-term dispersal (ie. colonisation) among sites according to three methods. The effective number of migrants between populations per generation ($N_e m$) was first estimated by conversion of multilocus pairwise F_{st} values in ARLEQUIN v3.5, where $F_{st} \approx 1/(4\ N_e m + 1)$. We also estimated $N_e m$ between sites according to the private allele method as implemented in GENEPOP v4.0.11. The private allele method does not involve explicit estimation of F_{st}, but it is based on the same population genetic models and the properties of allele frequency distributions that correspond to F_{st} (Slatkin 1985). Lastly, we calculated ratios of migration rate to mutation rate (M) and θ ($4\ N_e \mu$, where N_e is effective population size and μ is mutation rate) for pairs of subpopulations and genetic clusters using the MCMC maximum-likelihood method implemented in MIGRATE v3.2.7 (Beerli & Felsenstein 1999, 2001). Separate runs were carried out using the Brownian motion model with reference to pairwise geographical distances among sample sites. No reference to geographical distances was used for genetic clusters. Search criteria were set to 10 short chains of 10,000 steps and three long chains of 100,000 steps. Searches were combined across all long chains, and an adaptive heating scheme with four temperatures (1, 1.5, 3, and 10000) was used. Values of $N_e m$ were calculated for each pairwise comparison by multiplying the Maximum Likelihood (ML) estimate of M by $\theta/4$. Estimates of net immigration (the total number of immigrants minus the total number of emigrants) were then compared to identify migration asymmetries among sites and clusters (Hänfling & Weetman 2006).

Results

Genetic variation and population structure

All of the twelve loci were polymorphic among the individuals sampled for this study, exhibiting between 3 and 20 alleles per locus. Four of the polymorphic loci (Cgo22, Cgo33, Cgo1033 and Cgo56) were significantly out of HWE after sequential Bonferroni correction and therefore were excluded from all analyses except for assignment tests conducted with BAYESASS v1.3. We found consistently high levels of heterozygosity across the study area (Hänfling & Brandl 1998; Knaepkens et al. 2002, 2004b; Hänfling & Weetman 2006), with a mean heterozygosity of 0.598 among sites (Table 1). We also found that heterozygosity was higher in downstream sites than in upstream sites (Table 1). Longitudinal position explained most of the variance in heterozygosity among sites ($r^2 = 0.705$, $P = 0.021$). Similar patterns of decline were observed in the frequency of private alleles at each site (Table 1).

Comparisons based on linearised F_{st} values provide evidence of substantial genetic differentiation across the study area (Table 2). Progressively larger values were recovered for pairs of sites with increasing distance, and significant values were found for most pairwise comparisons of sites separated by a cascade upstream of Site F. The Mantel's test recovered a pattern of very strong IBD across the study area, where F_{st} values declined almost linearly with increasing distance between sites (Fig. 2, $r^2 = 0.844$, $P < 0.0001$). Autocorrelation of F_{st} values was significantly positive for sites <0.7 km apart ($P = 0.0191$) and significantly negative for sites more than 5.0 km apart (Fig. 2, $P = 0.0028$).

Neighbour-joining and Bayesian analyses recovered several distinct genetic clusters among the eight sample locations. The bootstrapped NJ tree topology (Fig. 3) based on D_{ps} values exhibited greatest support for nodes grouping sites [AB], sites [ABC] and [FGH]. Lesser support was recovered for nodes grouping [GH] and [DE]. Log-likelihood tests of P(X | K) values from STRUCTURE v2.1 runs recovered greater support for three clusters instead of two or four or more (all tests, $P < 0.0001$). Similarly, BAYESASS v1.3 simulations found that eight populations provided a poor fit to the data (i.e., where posterior probabilities were not significantly different from prior probabilities). Simulations involving three clusters produced the best fit (all tests, $P < 0.0001$). Both approaches grouped the study sites as [A B C], [D E] and [F G H]. The

Table 2. Pairwise statistics by site and cluster.

Site	A	B	C	D	E	F	G	H	Cluster	ABC	DE	FGH
A									ABC			
B	0.003								DE	**0.006**		
C	0.008	0.006							FGH	**0.043**	**0.029**	
D	**0.018**	0.013	0.000									
E	**0.021**	0.004	0.000	0.003								
F	**0.039**	**0.025**	**0.027**	**0.018**	0.009							
G	**0.071**	**0.055**	**0.056**	**0.045**	**0.039**	0.007						
H	**0.059**	**0.051**	**0.053**	**0.043**	**0.041**	**0.014**	0.005					

Site	A	B	C	D	E	F	G	H	Cluster	ABC	DE	FGH
A		5.4	4.1	3.4	2.4	1.4	1.0	1.3	ABC		9.0	1.2
B	Inf		3.9	3.3	2.9	2.1	1.6	1.9	DE	42.0		2.3
C	45.5	42.7		5.2	4.5	1.9	1.5	2.3	FGH	5.8	8.7	
D	19.4	22.6	Inf		5.5	3.1	2.1	2.3				
E	16.9	158.7	Inf	83.7		2.63	1.5	2.0				
F	8.3	11.2	11.8	15.5	50.5		6.0	6.0				
G	4.3	5.4	5.1	6.4	8.2	41.4		3.6				
H	5.3	5.7	5.5	6.7	7.8	21.9	70.9					

Inf = migration estimate not possible.
(Top) Pairwise F_{st} values. Bold F_{st} values significant at $P < 0.05$ after Bonferroni correction. (Bottom) Pairwise per generation number of migrants among sites using F_{st} (below diagonal) and private alleles (above diagonal).

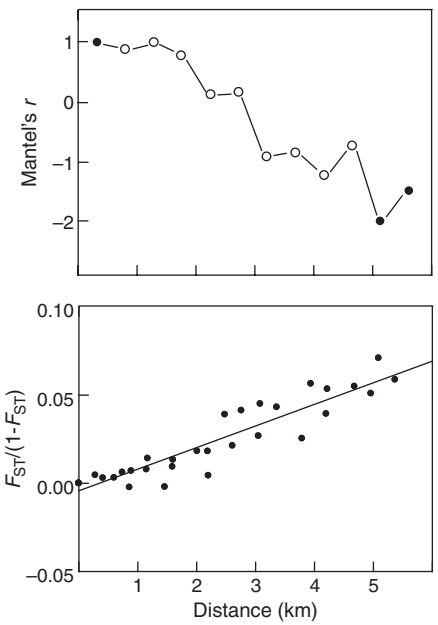

Fig. 2. (Top) Correlogram of Mantel's r versus linear distance across sample sites in the Nantahala River. Mantel's r values reflect autocorrelation of F_{st} values. Black circles are significantly different from zero. (Bottom) Linear regression of genetic distances and geographical distances among sample sites.

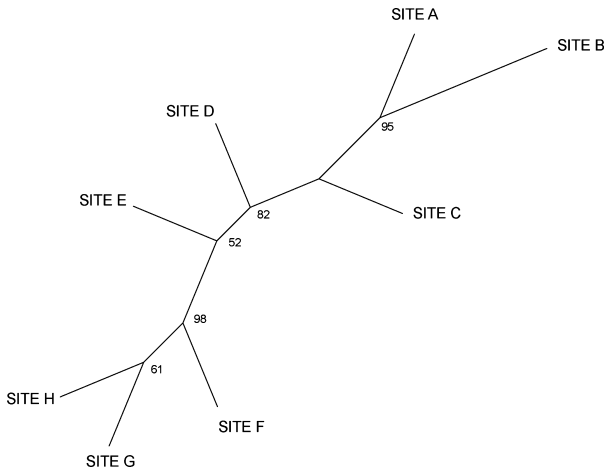

Fig. 3. Unrooted Neighbor-Joining tree of sampled populations based on pair-wise D_{ps} values among sites, with bootstrap support values >50%.

recovered patterns of genetic differentiation appear to reflect site proximity and the location of strong cascades more than other potential in-stream barriers, including pools and small dams (Fig. 1).

Patterns of variation among genetic clusters paralleled patterns observed among sites. Downstream clusters possessed greater heterozygosity and more private alleles than upstream clusters (Table 1). An AMOVA of the three clusters recovered a global F_{st} value of 0.028 ($P < 0.001$) and indicated that approximately 2.6% of genetic variance is attributable to among group differences ($P < 0.001$). Although comparable to those recovered in comparisons of sites, pairwise F_{st} values among clusters were all significant (Table 2); values ranged from 0.006 for the ABC–DE, to 0.029 for the DE–FGH comparison, to 0.043 for the ABC–FGH comparison.

Migration among populations

BAYESASS v3.1 nonequilibrium assignment tests placed between 68% and 95% of individuals to site of

capture, and from 93% to 97% of sampled individuals to cluster of capture (Table 3). The difference in assignment values between sites and clusters likely reflects the increase in statistical power that comes from having larger sample sizes for pairwise comparisons, and assignment of individuals to genetically definable subpopulations versus assignment of individuals to *a priori* defined locations. Most assignment values to sites other than those of capture were not significantly different from zero; though, nearly all individuals likely to be first-generation migrants were assigned to locations within the same genetic cluster as the location of capture (Table 3). Mean assignments by cluster resulted in nearly all individuals being assigned to their cluster of capture. Estimates of migration rates between adjacent clusters ranged from 0.01 to 0.06, with downstream estimates being about twice as large as upstream estimates (Table 3). The migration rates for downstream movement from [A B C] to [D E] and from [D E] to [F G H] were both significantly different from zero, where migration rates for upstream movement were not. STRUCTURE v2.1 equilibrium-based assignment tests produced similar but less discrete results.

Estimates of long-term $N_e m$ based on F_{st} values (Slatkin 1985) ranged from 4 for the most distinct pairwise comparison among sample sites (Site A vs. Site G) to infinity for comparisons among proximate sites (Table 2). Estimates of $N_e m$ between sites based on private alleles, on the other hand, ranged from 1 to 6 (Table 2). Among clusters, estimates of $N_e m$ based on F_{st} values ranged from 6 to 42 and estimates based on private alleles ranged from approximately 1–9. Pairwise ML values of $N_e m$ estimated by MIGRATE ranged from 0.0001 to 6.24 between sites, and pairwise ML values between clusters ranged from 0.66 to 0.87. A distinct migration asymmetry was found among sites based on net immigration values, with upstream Site A serving as a net sink of immigrants and all but one other site serving as net sources (Fig. 4). No asymmetry was found among genetic clusters (Fig. 4).

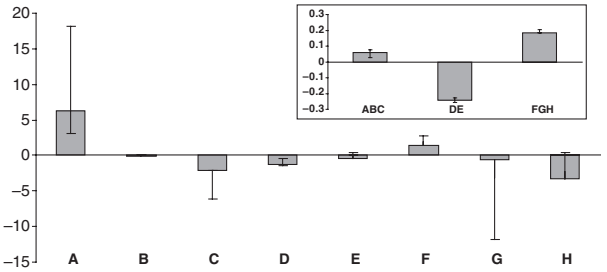

Fig. 4. Maximum likelihood based estimates of net immigration among sites and (inset) clusters with 95% confidence intervals. Positive values indicate that a site is a net recipient of immigrants. Negative values indicate that a site is a net source of immigrants.

Discussion

Distribution of genetic diversity

Variation in heterozygosity reflected the longitudinal position of sample locations in the study area, with heterozygosity higher in downstream sites than upstream sites. Evidence that upstream sites largely harbour subsets of alleles found in downstream sites and the near-absence of private alleles in upstream sites both indicate that genetic diversity increases with longitudinal downstream progression within the study area. Finding a similar pattern of declining genetic diversity among bullhead (*Cottus gobio*) from sites in the River Rye basin (England), Hänfling & Weetman (2006) suggested that genetic diversity can be predicted based upon longitudinal location of a population within a watershed. Our findings indicate that predictions can also be made at finer spatial scales, such as among stream segments separated by in-stream barriers. Work by Knaepkens et al. (2004b), showing no change in genetic diversity among bullhead sampled within a section of the Zwanebeek River dissected by a potentially impassable 35 m culvert, indicates that the limit to resolving spatial patterns of genetic diversity in sculpin is larger than a scale of tens of metres. Exceptionally predictable longitudinal patterns and strong relationships between genetic

Table 3. Bayesian assignments of individuals to each site and genetic cluster.

Site	A	B	C	D	E	F	G	H	Cluster	ABC	DE	FGH
A	**0.676**	0.004	**0.299**	0.004	0.004	0.004	0.004	0.006	ABC	**0.964**	0.032	0.0034
B	0.003	**0.675**	**0.296**	0.013	0.003	0.003	0.003	0.004	DE	**0.06**	**0.931**	0.01
C	0.003	0.004	**0.835**	**0.143**	0.004	0.003	0.003	0.004	FGH	0.006	**0.02**	**0.974**
D	0.003	0.003	0.043	**0.940**	0.003	0.003	0.002	0.004				
E	0.004	0.005	0.015	**0.270**	**0.677**	0.004	0.004	0.021				
F	0.003	0.003	0.004	0.010	0.003	**0.675**	0.003	**0.299**				
G	0.004	0.003	0.004	0.004	0.003	0.003	**0.675**	**0.304**				
H	0.003	0.003	0.007	0.031	0.004	0.003	0.003	**0.947**				

Values represent fraction of assignments to each cluster (±95% confidence interval values available upon request). Diagonal values represent assignment to capture location. Values below diagonal correspond to downstream migration (e.g., cluster ABC to cluster DE), and those above diagonal correspond to upstream migration (e.g., cluster DE to cluster ABC). Values in bold are significantly different from 0 at $P < 0.05$.

diversity, population size and physical condition factors (Knaepkens et al. 2002, 2004a) raise the possibility that genetic measures of sculpin populations can serve as useful indicators of stream connectivity and degradation, particularly for river-stream networks embedded within human-dominated landscapes (Bagley et al. 2002; Waits et al. 2008).

Historical colonisation versus contemporary dispersal

A longitudinal increase in genetic diversity as one moves downstream within a river-stream network could result from historical patterns of colonisation or contemporary dispersal reflecting movement bias, fragmented habitat or the presence of dispersal barriers (Hänfling & Weetman 2006; Morrissey & de Kerckhove 2009; DeHais et al. 2010). Discriminating between historical and contemporary factors is difficult, but evidence of migration–drift equilibrium can be a useful indicator of relative importance (Hutchison & Templeton 1999). For example, by testing for IBD (a condition reflecting migration-drift equilibrium) across increasing distance classes, Hänfling & Weetman (2006) inferred that patterns across the River Rye basin involving pairwise distances <15 km corresponded to contemporary conditions while patterns involving pairwise distances >30 km were attributable to historical factors. The observed pattern of strong IBD indicates that mottled sculpin are at migration–drift equilibrium in our study area, which is consistent with observations of population stability over comparable spatial scales (Freeman et al. 1988; Grossman et al. 1998; Petty & Grossman 2004). Therefore, longitudinal patterns of genetic diversity and long-term estimates of net immigration likely indicate that downstream sites have acted as a source for colonisation of upstream populations across evolutionary time (Slatkin 1985, 1993), while Bayesian estimates of downstream-biased dispersal instead reflect contemporary conditions (Hänfling & Weetman 2006).

The pairwise relationship between geographical and genetic distances observed in our study area is remarkably strong (Fig. 2), particularly considering that observations were made over such a fine spatial scale (for comparisons, see the study by Schweiger et al. 2004; Carlsson & Nilsson 2000; Vandewoestijne & Baguette 2004). Although a 'rule of thumb' holds that one or more migrants per generation results in genetic homogeneity, Wright (1931) and others (reviewed by Miller & Allendorf 1996) have shown that relatively low levels of exchange like those measured among our study sites (Table 2) and genetically defined clusters (Table 3) allow for substantial divergence in allele frequencies among subpopulations. Discovery of strong genetic differentiation across such short geographical distances would be unusual for less sedentary freshwater fish, including many economically important fishery and game species (Bunn & Hughes 1997). Studies of other sculpins (i.e., bullhead, *C. gobio*), however, have detected patterns of fine-scale population genetic structure (Hänfling & Weetman 2006). Many other species that exhibit strong IBD over fine spatial scales also have limited dispersal capacity (e.g., snails, Schweiger et al. 2004; Arnaud et al. 1999; amphibians, Storfer 1999; Larson et al. 1984; stoneflies, Schultheis et al. 2002), although at least one exception exists (e.g., brown trout, Carlsson & Nilsson 2000).

While distance is a key factor generating genetic structure among the sample sites, in-stream barriers also influenced patterns of genetic variation and dispersal. Relatively high values of F_{st} were found among sites within the study area, particularly among sites separated by cascades. Neighbour-joining (Fig. 3) and Bayesian estimates of population structure also indicated that the study sites correspond to genetic clusters that reflect site proximity and the location of strong cascades (Fig. 1). Bayesian estimates of migration among clusters suggest that individuals are either actively moving downstream or are being carried downstream across cascades. Cascades likely act as physical and energetic barriers to upstream movement in the study area, In addition, pools at the base of cascades may function as ecological traps, as they often harbour piscivorous trout (Pulliam 1988; Weldon & Haddad 2005).

All of the approaches we used to estimate dispersal generated consistently low values of migration. Concordance among genetic estimators is generally considered to be among the best criterion of robustness (Slatkin & Barton 1989). Finding general agreement among dispersal estimates derived from distinct genetic methods for mottled sculpin contrasts with other studies that have recovered inconsistent estimates for other stream fish (Wilson et al. 2004). This may be due to the strong population genetic differentiation and conditions approximating migration–drift equilibrium within our study area (Slatkin & Barton 1989; Whitlock & McCauley 1999; Wilson et al. 2004).

Comparison between genetic and mark–recapture estimates of dispersal

Genetic and mark–recapture approaches yield distinct but consistent and complementary perspectives on dispersal rates and directional bias in mottled sculpin. As most differences reflect the spatial and temporal resolution of each approach, combining information from both approaches can offer a more comprehensive assessment of mottled sculpin movement and popula-

tion connectivity than would an assessment based on only one approach.

Genetic estimates of among-site contemporary migration and maximum movement potential are somewhat greater than those reported by mark–recapture studies. Bayesian estimates, for example, suggest that between 1% and 6% of individuals are recent migrants between adjacent genetic clusters. A mark–recapture study by Breen et al. (2009) found that 16% of moves exceed 100 m per year and that 3% of moves exceed 200 m or more. The observed positive autocorrelation of genetic structure up to 700 m apart (Fig. 2) and estimates of exchange among clusters separated by >800 m suggest that moves of this extent occur in the study area with some regularity. Although comparable to the maximum movement distance (511 m) ever recorded (Breen et al. 2009), these estimates exceed the longest move (165 m) observed in most other studies by 400% or more (Bailey 1952; Gerking 1959; MacCleave 1964; Brown & Downhower 1982; Hill & Grossman 1987; Grossman et al. 1998; Larson et al. 2002; Petty & Grossman 2004; Lamphere 2005; Breen et al. 2009; Petty & Grossman 2010).

Several factors could explain the observed discrepancies between genetic and mark–recapture estimates of movement potential. The inferred movements of 700 m or more may be the result of shorter (e.g., 100 m) movements integrated over time via reproduction among neighbouring individuals (Dobson 1994). Alternatively, the inferred movements of 700 m or more may the result of longer moves, like those reported by Breen et al. (2009) integrated across a full lifespan which can be as long as 7 years or more (Grossman et al. 2002). Differences might also reflect movement during early life stages that is not easily captured by mark–recapture approaches (Morrissey & Ferguson 2011), or the well-documented tendency of mark–recapture studies to underestimate dispersal extent at the population level (reviewed by Koenig et al. 1996; Gowan et al. 1994).

Although potentially complicating, incongruent estimates of dispersal can provide more thorough understanding of movement and population connectivity. In mottled sculpin, inferences about historical patterns of colonisation can be drawn from the genetic structure of populations, while estimates of contemporary dispersal over large and small spatial scales can be derived from genetic assignment tests and mark–recapture studies. For example, mark–recapture studies have found an upstream bias to movement (Lamphere 2005; Breen et al. 2009), whereas assignment tests of recent migrants indicate that contemporary dispersal occurs in a downstream direction. These findings do not necessary conflict, as they may reflect the presence or absence of in-stream barriers, and the direction of dispersal according to different spatial or temporal scales. Over relatively small time periods (i.e., days to months), mottled sculpin may tend to move upstream, with individuals accumulating at the downstream end of barriers. The presence of barriers could result in net downstream dispersal over similar time periods, yet infrequent colonisation of upstream reaches could still occur as barriers fail over longer time scales (i.e., years to decades and beyond). Genetic and demographic methods may also be providing information on movement at different life-history stages. Demographic approaches therefore could be measuring adult movement behaviour, while genetic approaches could be measuring movement corresponding to early life stages or lifetime movement (Morrissey & Ferguson 2011).

Conclusions

A general consensus is emerging that a combination of demographic and genetic approaches provides more valuable information on dispersal than either approach can alone (Wilson et al. 2004; Howeth et al. 2008; Lowe et al. 2008). Direct and indirect methods can provide consistent information on dispersal, as has been shown in studies of fish and other aquatic species where both methods depict similar patterns (Chenuil et al. 2000; Newman & Squire 2001; Adams & Hutchings 2003; Heggenes et al. 2006; Howeth et al. 2008; Lowe et al. 2008). Our findings indicate that mark–recapture and genetic methods can consistently depict many aspects of individual movement. Our results also provide an example of how incongruencies and discrepancies can offer valuable information that might not otherwise come to light. It is important to note, however, that inferences drawn from a combination of approaches should be viewed as hypotheses that warrant further testing through experimentation or spatially explicit models that have so far relied on mark–recapture data to understand demographic implications of individual behaviour (DeAngelis & Gross 1992; Rashleigh & Grossman 2005).

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